NeuroCardia

An Open-Source, On-Device Phase-Domain Diagnostic Engine for Single-Lead ECG

White Paper — Technical and Regulatory Overview

Version 0.2 — Revision

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October 14, 2026

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Revision history. This is revision 0.2 of the NeuroCardia White Paper. Revision 0.1 (October 14, 2026) reported the pilot results of the Kuramoto residual (AUC 0.89 on a paired corpus of nine sinus and nine AF recordings) and the trajectory-correlation outcome of the five-arm sensitivity analysis. Revision 0.2 adds: the residual AUC for all five experimental arms (two of which do not exhibit the residual); the orthogonality result (the residual retains AUC 0.716 after regressing out $\mathrm{sd1}/\mathrm{sd2}$ and the coefficient of variation); the finding that trajectory robustness and residual robustness are disjoint; and a scope condition on the residual's frequency-distribution dependence. The regulatory strategy is unchanged. The list of claims requiring additional validation now includes residual orthogonality to standard HRV.

Abstract

NeuroCardia is a free, open-source Android application for offline, on-device analysis of single-lead electrocardiogram (ECG) recordings. It requires no network connection, no user accounts, and no telemetry; every computation runs on the handset, and the application functions identically in airplane mode. The application computes conventional heart rate variability (HRV) metrics, a four-axis signal quality score, and a phase-domain Cardiac Synchronization Index (CSI) derived from a Kuramoto coupled-oscillator network driven by the RR tachogram. Its DSP and I/O layers are packaged as standalone Kotlin/JVM modules that can be reused in any JVM project.

This document specifies the computational model, the diagnostic metrics, the validation strategy against public reference datasets, and a preliminary regulatory pathway toward FDA clearance as a Class II medical device under 510(k). It is explicit about which claims are literature-supported, which are novel and awaiting validation, and which are research hypotheses. The CSI is presented as a research feature, not a clinical diagnostic; the application's primary contribution is the deployment envelope — open-source, offline, on-device ECG analysis on commodity Android hardware. The source code is released under GPL-3.0-or-later.

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1 Introduction

1.1 Motivation

Heart rate variability analysis has been clinically accepted for decades as a window into autonomic function [1]. The algorithms are mature and well-understood. What is not widely available is a *deployment envelope* in which those algorithms can be run without a workstation, without a network connection, and without sending patient data to a third party.

The dominant open-source tools for ECG and HRV analysis — predominantly Python-based research stacks — assume a workstation, a network connection, and a user comfortable with a command line. Field clinicians, rural health workers, air-gapped research environments, and privacy-sensitive deployments are left without validated, offline alternatives. Commercial Holter analysis software is closed-source and typically tied to a specific vendor's hardware. The gap is not in the algorithms; it is in the delivery.

NeuroCardia addresses that gap with four design commitments.

1. **On-device only.** No cloud, no accounts, no telemetry. The application functions identically in airplane mode. Every computation, from R-peak detection to diagnostic report generation, runs on the handset.

2. **Open source.** GPL-3.0-or-later. Every algorithm is auditable, reproducible, and forkable. There is no proprietary binary blob and no vendor lock-in.

3. **Reusable modules.** The DSP and I/O layers are packaged as pure-Kotlin/JVM libraries with no Android dependency. They can be consumed by any JVM project and tested in under a second, independent of the Android application that uses them.

4. **Standard metrics first, novel metrics clearly labeled.** Time-domain and frequency-domain HRV metrics are computed using standard formulations and are the application's validated outputs. The phase-domain Cardiac Synchronization Index is presented as a research feature whose record-level diagnostic behavior is documented in the companion residual paper and whose clinical utility is not yet established.

The primary contribution of the project is not a new algorithm. It is the combination of standard algorithms with a deployment envelope that does not require network access, third-party services, or a specific hardware vendor.

1.2 Scope of this document

This white paper specifies: the computational model (Section 3), the diagnostic metrics and their intended clinical meaning (Section 4), the validation strategy against public datasets (Section 5), the regulatory pathway toward FDA clearance (Section 6), the limitations and open questions (Section 7), and the development roadmap (Section 8).

It is written for three audiences: clinical collaborators evaluating the diagnostic claims, regulatory reviewers assessing the device pathway, and software engineers who may contribute to the implementation.

1.3 What this document is not

This is not a clinical validation study. It does not report prospective trial data, and it makes no claim of diagnostic accuracy against a gold standard. It is a design and rationale document. Validation is described as a strategy (Section 5), not as a result.

1.4 What this application is not

NeuroCardia is not a replacement for standard ECG interpretation, and it is not a diagnostic device. Three specific exclusions are stated here because they shape every subsequent claim.

- **It does not diagnose.** It computes descriptive metrics from an ECG recording. It does not produce a diagnosis, does not recommend treatment, and does not triage patients. Every report produced by the application includes a disclaimer that the output is informational and requires review by a qualified clinician.
- **The CSI is a research feature.** The Cardiac Synchronization Index and its derived early warning signals are not validated diagnostics. They are documented in this paper and in the companion residual paper as record-level descriptors whose clinical significance is a subject of ongoing investigation. They are gated behind a settings flag and labeled as not-for-clinical-use.
- **It is not a real-time monitor.** The application analyzes completed ECG recordings loaded from files. It does not stream from a device, does not monitor continuously, and does not alarm. The phase-domain metric in particular requires the full recording to stabilize and is not a short-window detector.

These exclusions are deliberate and are not limitations of the current version. They define the intended use. A device that made any of these three claims would be a different device with a different regulatory pathway and a different validation burden.

2 Background

2.1 Heart rate variability

HRV analysis quantifies beat-to-beat variations in the RR interval series. Standard metrics include time-domain measures such as the root mean square of successive differences (RMSSD), the standard deviation of normal-to-normal intervals (SDNN), and the proportion of successive differences greater than 50 ms (pNN50); and frequency-domain measures such as low-frequency power (LF, 0.04–0.15 Hz), high-frequency power (HF, 0.15–0.40 Hz), and the LF/HF ratio [1].

These metrics are descriptive: they characterize the state of the autonomic nervous system at a moment in time. They do not, by themselves, describe how that state is evolving, and they do not surface the dynamical signatures of impending transitions.

2.2 Cardiac rhythm as a coupled oscillator system

The sinoatrial node is a population of coupled pacemaker cells whose collective firing produces the cardiac rhythm. This is mathematically a system of coupled phase oscillators, and it is naturally described by the Kuramoto model [2, 3]:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \tag{1}$$

where θ_i is the phase of oscillator i, ω_i its natural frequency, K the coupling strength, and N the number of oscillators. The degree of synchronization is captured by the order parameter:

$$r = \left|\frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j}\right|, \tag{2}$$

with $r \to 1$ indicating full phase locking and $r \to 0$ indicating incoherence.

2.3 Prior work on phase analysis of cardiac rhythm

Phase synchronization indices applied to ECG have been shown to outperform correlation, coherence, and mutual information for detecting rotors during atrial fibrillation. Early warning signals derived from Kuramoto-type models have been demonstrated to detect impending transitions to fibrillation in computational studies. What is missing is an integrated, open-source, on-device implementation.

3 Computational Model

3.1 RR-to-phase mapping

Given an RR interval series $\{R_k\}_{k=1}^{M}$ (in seconds), we construct a phase trajectory by cumulative integration:

$$\phi_k = 2\pi\sum_{j=1}^{k}\frac{\overline{R}}{R_j} \pmod{2\pi}, \tag{3}$$

where $\overline{R}$ is the mean RR interval. Each beat advances the phase by $2\pi\overline{R}/R_j$, so a beat at the mean rate advances exactly 2π. This maps the irregular cardiac rhythm onto a unit circle and preserves the relative timing of each beat.

3.2 Network construction

We instantiate a Kuramoto network of $N = 16$ oscillators. Natural frequencies are assigned deterministically from the non-trivial Riemann zeros γ_i, then normalized to zero mean and unit variance:

$$\omega_i = \frac{\gamma_i - \bar{\gamma}}{\sigma_\gamma}, \tag{4}$$

where $\bar{\gamma}$ and σ_γ are the mean and standard deviation of the first 16 Riemann zeros. The normalization is required for numerical tractability: the reference parameterization places the carrier at $\Omega_T = 587{,}000$ rad/s, five orders of magnitude above any frequency the Euler integrator can resolve at a practical time step.

The frequency distribution is a specific modeling choice. Section 5.4 reports the sensitivity of the diagnostic to this choice. The full sensitivity analysis is in the companion document [10].

The coupling strength is $K = 2$. This value is not arbitrary: as shown in Section 5.4, the CSI at $K = 2$ carries incremental diagnostic information over conventional HRV, while at higher coupling strengths the network saturates and the metric becomes a monotone function of RR variability.

3.3 Phase reset from beats

At each detected R-peak, we reset the phase of a dedicated heart oscillator (index 0) to the cumulative phase ϕ_k from Equation 3. The remaining $N - 1$ oscillators evolve under Equation 1. This reset is what couples the model to the actual cardiac signal.

3.4 Coupling topologies

Four coupling topologies are supported, selectable from a settings menu: all-to-all (default), nearest-neighbor, small-world, and star. The topology is a modeling choice, not a diagnostic one.

3.5 Order parameter trajectory

We integrate the network with a fixed step $\Delta t = 0.05$ s and record the order parameter r at each step. The result is a time series $r(t)$ that we term the Cardiac Synchronization Index (CSI).

4 Diagnostic Metrics

4.1 Cardiac synchronization index

The CSI $r(t)$ is the primary novel output. It is bounded in $[0, 1]$ by construction. In the production configuration ($K = 2$), the mean value across the reference corpora falls in the range 0.60–0.70; at higher coupling strengths the metric saturates and sits at 0.98–0.99 for all rhythms. The interpretation of specific CSI values is therefore not fixed across parameterizations, and the metric should be read as a relative quantity within a fixed parameterization, not as an absolute one.

The trajectory of $r(t)$ — its derivative, variance, and higher moments — is as diagnostic as its instantaneous value. The app reports both the instantaneous value and the trajectory statistics.

The CSI is a parameterized quantity. Its value depends on the frequency distribution of the oscillator network, the coupling strength K, and the number of oscillators N. It is not a universal invariant of cardiac phase synchronization.

4.2 Early warning signals

At bifurcation points in the Kuramoto model, the statistical moments of the order parameter change predictably. We compute four EWS metrics over a sliding window: variance, lag-1 autocorrelation, skewness, and kurtosis. These metrics are displayed as a composite risk strip below the main CSI plot, flagging potential transitions before they manifest as overt arrhythmias. The EWS metrics are exploratory; their prognostic value has not been prospectively validated.

4.3 Respiratory sinus arrhythmia and vagal coherence

We extract a respiration proxy from ECG baseline wander in the 0.1–0.4 Hz band, then compute coherence between RSA and the high-frequency component of the RR spectrum. This is the vagal-cardiac coherence index. It is a proxy for the brain-heart axis derived from single-lead ECG alone.

4.4 Apnea screening via coherence band tracking

Following the reported literature on heart-respiration coherence in sleep apnea, we track the coherence peak in the 0.2–0.3 Hz band over time. A vanishing peak is associated with apneic episodes. We report a per-window apnea likelihood score. This is explicitly a screening aid, not a diagnosis.

4.5 Signal quality and runtime verification

Every metric in this section is only as trustworthy as the signal from which it was derived. NeuroCardia therefore computes a signal quality assessment on the raw input, at load time, before any detection or analysis runs. Four independent sub-metrics are computed:

1. Signal-to-noise ratio (SNR). The ratio of energy in the QRS band (5–15 Hz) to energy in the high-frequency noise band (40–100 Hz). Reported in decibels.
2. Flatline fraction. The proportion of successive sample pairs whose absolute difference is below an epsilon.
3. Clipping fraction. The proportion of samples at the observed extremes of the recording.
4. Baseline wander fraction. The ratio of energy below 0.5 Hz to total energy.

These four are combined into a composite score in $[0, 1]$. Hard floors apply: if the flatline fraction exceeds 0.9, or the baseline wander fraction exceeds 0.7, or the SNR is below 0 dB, the composite score is capped regardless of the other terms. The score maps to a verdict: GOOD $\geq 0.75 >$ FAIR $\geq 0.50 >$ POOR. The verdict is displayed in the analyzer screen and included in every report.

This is a runtime verification mechanism, not a validation claim. It asserts that the input from which the metric was derived meets a minimum quality threshold, not that the metric is correct.

5 Validation Strategy

5.1 Reference datasets

To establish analytical validity before clinical trials, NeuroCardia is tested against three gold-standard annotated corpora from PhysioNet:

- **MIT-BIH Arrhythmia Database** [5]. 48 half-hour records of ambulatory ECG, annotated beat-by-beat by cardiologists.
- **Fantasia Database**. 40 young and elderly healthy subjects with rigorously screened 2-hour records.
- **European ST-T Database**. 90 two-hour recordings focused on ST and T-wave changes.

5.2 Validation pipeline

Validation runs as an automated test harness in the CI/CD pipeline. For every commit: records from the benchmark corpora are streamed through the app's Kotlin data ingestion pipeline; R-peak annotations produced by the Pan-Tompkins module are compared against reference `.atr` annotation files using a 150 ms match window; the Kuramoto engine computes $r(t)$ for each record; and summary statistics are exported to CSV and verified against reference golden outputs.

5.3 Acceptance criteria

Each validation test has an explicit numeric target. The targets below are the commitments made in this white paper.

The residual orthogonality criterion is new in revision 0.2. It asks whether the Kuramoto residual retains diagnostic value after regressing out the two standard HRV features it is most correlated with. The current implementation meets the criterion (AUC 0.716 at $n = 18$), but the margin is narrow and the sample size small.

Metric	Dataset	Target
R-peak sensitivity	MIT-BIH	$\geq 99.0\%$
R-peak PPV	MIT-BIH	$\geq 99.0\%$
HRV (RMSSD) agreement	Fantasia	within 5%
HRV (SDNN) agreement	Fantasia	within 5%
CSI discriminative validity ($K = 2$)	NSR/AF pilot	AUC ≥ 0.85
Residual orthogonality	NSR/AF pilot	AUC ≥ 0.65 after HRV removal
EWS prognostic validity	MIT-BIH AF transitions	AUC ≥ 0.70
Signal quality screen	All datasets	score ≥ 0.5

Table 1: Acceptance criteria. All targets are met by the current implementation with the exception of the EWS prognostic criterion, which is exploratory and has not yet been tested on a corpus with annotated transitions.

5.4 Empirical results

The CSI depends on two parameters that are not determined by cardiac physiology: the frequency distribution of the oscillator network, and the coupling strength K. To characterize this dependence and to test whether the CSI carries information beyond conventional HRV, we ran a series of experiments against two reference corpora: the MIT-BIH Arrhythmia Database (48 records, mixed rhythms) for the parameter sensitivity analysis, and a paired sample of MIT-BIH Normal Sinus Rhythm (9 records) and MIT-BIH Atrial Fibrillation (9 records) for the discriminative validity test.

5.4.1 Sensitivity to the frequency distribution

Five frequency distributions were tested: Riemann zeros (Arm A, the production configuration), uniform random (Arm B), Gaussian (Arm C), physiological scale (Arm D), and linear spacing (Arm E). Table 2 reports the mean Pearson correlation between Arm A's trajectory and each alternative arm's trajectory.

K	Arm B	Arm C	Arm D	Arm E
2.0	0.052	0.076	0.051	0.168
5.0	0.250	0.230	0.250	0.611
10.0	0.386	0.366	0.386	0.778
30.0	0.458	0.428	0.458	**0.869**
100.0	−0.000	0.000	0.001	0.001

Table 2: Mean Pearson correlation between the CSI trajectory under Arm A and under each alternative arm, across 48 mitdb records. The pre-registered threshold of 0.80 is met only by Arm E at $K = 30$, where the network is saturated.

The CSI trajectory is not robust to the frequency distribution at any coupling strength where the metric has dynamic range. The companion sensitivity analysis reports the full result.

5.4.2 Sensitivity to coupling strength

The coupling strength was swept over $K \in \{0.5, 1, 2, 3, 4, 5, 10\}$. The residual diagnostic peaks at $K = 2$ (AUC 0.89), inverts below $K = 1$ (AUC 0.17–0.21), and decays to noise above $K = 3$ (AUC 0.59–0.72). The useful window is narrow.

5.4.3 Correlation with conventional HRV

At $K = 10$, the mean CSI correlates with the coefficient of variation of the RR interval series at $\rho = -0.95$, with SDNN at $\rho = -0.87$, and with the Poincaré SD2 at $\rho = -0.86$. The CSI is essentially a monotone function of RR variability at this parameterization.

At $K = 2$, the CSI is not a monotone function of RR variability, and the residual (the component orthogonal to RR variability) separates sinus from AF.

5.4.4 Residual analysis

We regressed CSI mean against RR coefficient of variation by ordinary least squares and analyzed the residuals. Table 3 reports the residual AUC at $K = 2$ for each of the five frequency distributions.

Arm	Distribution	Residual AUC	p
A	Riemann zeros	0.889	0.002
B	Uniform random	0.543	0.986
C	Gaussian	0.852	0.007
D	Physiological scale	0.556	0.982
E	Linear spacing	0.840	0.010

Table 3: Residual AUC at $K = 2$ for each frequency distribution, on the paired sinus/AF pilot corpus. Three arms exhibit the residual; two do not.

The residual is diagnostic for three of five tested distributions. Revision 0.1 of this white paper reported only the three working arms and described the result as robust across the frequency distribution; the full five-arm test shows that this claim is narrower than reported.

5.4.5 Orthogonality of the residual to standard HRV

To test whether the residual carries information beyond standard HRV, we regressed it on the two features it is most correlated with in the pilot corpus (the Poincaré SD1/SD2 ratio and the coefficient of variation of the RR series) and computed the AUC of the orthogonalized residual. The raw residual separates sinus from AF with AUC 0.889; the orthogonalized residual separates them with AUC 0.716. Approximately two thirds of the residual's diagnostic value is attributable to its linear relationship with the two HRV features; one third is not.

At $n = 18$ the confidence interval on the orthogonalized AUC is wide and the difference between 0.889 and 0.716 is not statistically significant at this sample size. The finding is suggestive, not conclusive.

5.4.6 Mechanism

In the AF cohort, 94% of the residual's *between-record* variance is explained by three P-wave features extracted from the raw ECG (P-wave energy, PR-interval variability, and fibrillatory amplitude), features the Kuramoto engine does not have access to. In the sinus cohort, 37% is explained. The residual is best understood as a between-record discriminator, not a within-record state tracker: the window-to-window variation of the residual within a single record does not correlate with the same features (mean $|r| < 0.15$ in both cohorts), and the window-level AUC plateaus at 0.66–0.68 across 1–5 minute windows (0.50 at 30 s). Whole-record averaging is required to recover the discriminative signal. Two thirds of the residual's record-level diagnostic value is redundant with standard HRV.

5.4.7 Interpretation

Three conclusions follow.

First, the CSI at $K = 2$ carries information about cardiac rhythm that is not captured by conventional HRV. After regressing out the linear dependence on RR variability, the residual separates normal sinus rhythm from atrial fibrillation with an AUC of 0.89 on the pilot corpus. After regressing out both RR variability and the Poincaré SD1/SD2 ratio, the residual retains an AUC of 0.716.

Second, the metric's behavior depends critically on the coupling strength. The same metric at $K = 10$ is redundant with conventional HRV, with no residual diagnostic information. Above $K = 3$ the network saturates; below $K = 1$ the sign of the residual correlation inverts. The useful regime is narrow.

Third, the residual is not distribution-agnostic. It is exhibited by three of five tested frequency distributions. The choice of distribution matters.

5.5 Reproducibility

All validation code and expected outputs are committed to the repository under `validation/`. Every metric is reproducible with a single Gradle command on any JVM:

```
./gradlew :validation:run --args="<mode> <args>"
```

The output is deterministic given the same input and the same build. Builds are signed, and the signed APK hash is published with every release.

6 Regulatory Pathway (FDA 510(k))

6.1 Intended use and device classification

NeuroCardia is intended to be used as a software-as-a-medical-device (SaMD) decision-support tool. It processes single-lead ECG data to compute heart rate variability metrics and phase-domain synchronization indices. It is intended for use by healthcare professionals and trained investigators in research and clinical settings. It does not provide an automated diagnosis, does not direct clinical treatment, and is not intended as a replacement for standard ECG interpretation.

Under FDA product classification, ECG analysis software of this type falls under product code DQK (electrocardiograph software) or MWJ (system, network, and patient monitoring), typically regulated as Class II medical devices requiring 510(k) clearance.

6.2 Predicate device selection

The primary predicate device for 510(k) clearance will be an established commercial ECG analysis software package with cleared indications for automated HRV analysis and arrhythmia detection from ambulatory ECG recordings.

6.3 Substantial equivalence argument

NeuroCardia is substantially equivalent in intended use and fundamental technological characteristics to predicate devices. The technological difference — the introduction of the Kuramoto coupled-oscillator model and the derived CSI $r(t)$ — represents a new feature for a cleared SaMD. Under FDA guidance, new analytical methods in an existing device category are addressed through analytical validation and non-clinical performance testing, without necessarily requiring a De Novo pathway, provided the device does not make novel diagnostic claims that alter its risk profile.

6.4 FDA classification strategy

We plan a 510(k) premarket notification pathway. The claims made in the initial submission are: R-peak detection and heart rate computation; HRV metrics (time and frequency domain); rhythm irregularity flagging; and report export (PDF) suitable for clinical documentation.

Claims that require additional validation before regulatory submission: cardiac synchronization index as a diagnostic; early warning signal detection; apnea screening; vagal-cardiac coherence index; and residual orthogonality to standard HRV.

These are initially released as research features, gated behind a settings flag, and documented as not-for-clinical-use. A subsequent 510(k) or De Novo submission will cover them once validation is complete.

6.5 Software lifecycle and quality management

To comply with FDA expectations for SaMD (and IEC 62304 standard for medical device software lifecycle processes), development adheres to version control and reproducibility (every release is tagged, cryptographically hashed, and tied to a Software Bill of Materials in CycloneDX format); traceability (requirements are mapped to unit tests and integration tests); risk management (ISO 14971; identified hazards such as false R-peak classification are mitigated through signal quality checks, confidence scoring, and mandatory clinician review disclaimers); and quality management (ISO 13485-aligned QMS before any clinical claim).

7 Limitations and Open Questions

7.1 The Riemann-zero frequency distribution

The author does not claim that the Riemann zeros have a direct physiological correlate in cardiac electrophysiology. The choice is a specific choice documented in the companion mathematical reference. Whether this distribution offers diagnostic advantages over standard Gaussian or uniform frequency distributions is addressed empirically in Section 5.4. The sensitivity analysis shows that the CSI trajectory is not invariant to the choice of frequency distribution at any tested coupling strength, and that the residual diagnostic is exhibited by three of five tested distributions.

7.2 Carrier scale and phase velocity

The reference parameterization places the carrier frequency at $\Omega_T = 587{,}000$ rad/s, five orders of magnitude above the cardiac rhythm. The absolute scale of the carrier does not affect the dimensionless order parameter $r(t)$, but it has a decisive effect on numerical integration. We work with normalized frequencies, in which the carrier is subtracted and the residual Riemann-zero distribution is scaled to unit variance.

7.3 Signal quality and artifact sensitivity

Like all HRV and phase-domain tools, NeuroCardia is sensitive to motion artifact, electrode displacement, and ectopic beats. While the Pan-Tompkins detector includes amplitude and slope filtering, severe baseline wander or frequent premature ventricular contractions can distort the RR tachogram.

7.4 Choice of $N = 16$

The number of oscillators is a free parameter. We adopt $N = 16$ to match the reference implementation. The sensitivity of the CSI to N is a planned study.

7.5 Single-lead constraints

The absence of a spatial map of the atrium limits rotor detection to a temporal analysis. A multi-lead device would allow spatial mapping. That is a different device with a different intended use.

7.6 Residual orthogonality

The orthogonality result (AUC 0.716 at $n = 18$) has a wide confidence interval. Full-corpus replication on the complete NSRDB and AFDB is required before the residual can be claimed as a validated discriminative feature.

7.7 Validation is ongoing

At the time of writing, the novel metrics are supported by literature and by internal consistency checks, but not by prospective clinical validation. This document will be revised as validation results become available. Readers should treat every claim in Section 4 as a hypothesis under test, not as an established result.

8 Roadmap and Future Work

Development milestones for NeuroCardia are structured across five version releases:

- **v0.1 (Current).** Core Android app, Pan-Tompkins R-peak detector, basic time-domain HRV, Kuramoto engine with $N = 16$ and Riemann zero frequencies, CSV (RR intervals) and PDF report export, GPL-3.0 open-source release.
- **v0.2.** EWS metrics (variance, autocorrelation), CSV exports for metrics and annotated signals, JSON audit log and SHA-256 manifest generation.
- **v0.3.** Single-file HTML reports, session bundle JSON schema (v1), automated SBOM generation.
- **v0.4.** Full $r(t)$ CSI trajectory export, advanced coupling topology selectors, companion UI plots.
- **v0.5.** WFDB (.dat + .hea + .atr) and EDF+ export support for direct interoperability with PhysioNet and clinical workstations.

9 Conclusion

NeuroCardia is an open-source Android application for offline, on-device analysis of single-lead ECG recordings. It runs entirely on the handset, with no network dependency, no user accounts, and no telemetry. It computes conventional HRV metrics, a four-axis signal quality score, and a phase-domain Cardiac Synchronization Index derived from a Kuramoto coupled-oscillator network. Its DSP and I/O layers are packaged as standalone Kotlin/JVM modules that can be reused in any JVM project.

The application's primary contribution is the deployment envelope: open-source, offline, on-device ECG analysis on commodity Android hardware, with a regulatory strategy that treats the conventional HRV metrics as validated outputs and the phase-domain metric as a research feature with a documented plan for further validation. The phase-domain metric itself is a record-level nonlinear summary of the RR series with partial novelty over standard HRV (approximately one third of its diagnostic value is not explained by the two HRV features it is most correlated with) and with specific, quantified limitations (three of five tested frequency distributions; record-level but not window-level discrimination). These properties are documented in the companion residual paper.

The version 0.1 white paper is revised here as version 0.2. The principal changes are a reframing of the application's contribution around the deployment envelope rather than the novel metric, the addition of an explicit "what this application is not" section, and the scoping of the phase-domain claims to their empirically demonstrated range. The regulatory strategy is unchanged.

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